

repo-to-pdf

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Changelog

v0.1.9(2026-02-22)

- upgrade puppeteer to get outline; add concurrency to puppeteer (#28)
- enable local file for wkhtmltopdf (#22)
- bump ci deps

v0.1.8(2023-08-01)

- fix whitelist filtering
- replace relaxedjs with raw puppeteer
- add a pdf renderer wkhtmltopdf, faster but has split lines issue

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repo-to-pdf

 v0.1.9 (<https://www.npmjs.com/package/repo-to-pdf>)

 105/month (<https://www.npmjs.com/package/repo-to-pdf>)

Make pdf from source code

Here is a [sample](https://github.com/josherich/repo-to-pdf/blob/master/sample.pdf) (<https://github.com/josherich/repo-to-pdf/blob/master/sample.pdf>)

Install

```
npm install repo-to-pdf
```

Usage

Basic

```
npx repo-to-pdf [src/folder]
```

```
const {generateEbook} = require('repo-to-pdf')

/**
 * @typedef Options
 * @type {object}
 * @property {string} renderer - [node|calibre|wkhtmltopdf] can be either node,
calibre and wkhtmltopdf
 * @property {string} calibrePath - path of calibre's ebook-convert
 * @property {string} pdf_size - pdf size limit, in bytes
 * @property {string} white_list - list of file extensions to be included,
separate by ','
 * @property {string} format - [mobi|epub|pdf] can be either mobi, epub, pdf
 * @property {string} device - [desktop|tablet|mobile] style can be opt for
desktop, tablet and mobile
 * @property {string} baseUrl - base url of CSS style files
 * @property {boolean} outline - include PDF outline/bookmarks (default true)
 * @property {number} concurrency - max number of parallel Puppeteer PDF jobs
(default auto)
 */

/**
 * Generate ebook from content folder with the given file format
 * @param {string} inputFolder - content folder
 * @param {string} outputFile - output file name
 * @param {string} title - title in ebook file
 * @param {Options} options
 */
generateEbook(
  './',
  'test.pdf',
  'repo-test',
  {renderer: 'node', pdf_size: 3*0.8*1000*1000, format: 'pdf', device: 'desktop',
outline: true}
)
```

Command Line Options

```
-d, --device [platform]
device [desktop(default)|mobile|tablet]
# npx repo-to-pdf ../repo -d mobile

-t, --title [name]
pdf filename
# npx repo-to-pdf ../repo -t MeinKampf.pdf

-w, --whitelist [wlist]
file format white list, split by ","
# npx repo-to-pdf ../repo -w js,md

-s, --size [size]
pdf file size limit, in MB, default 10 MB
# npx repo-to-pdf ../repo -s 10

-r, --renderer [node|calibre|wkhtmltopdf]
use either node(relaxedjs) or calibre to render ebook, node outputs pdf, calibre
outputs pdf, mobi, epub
# npx repo-to-pdf ../repo -r calibre

-f, --format [pdf|mobi|epub]
output format, either pdf, mobi, epub. mobi and epub are generated using calibre
ebook-convert
# npx repo-to-pdf ../repo -f mobi

--no-outline
disable PDF outline/bookmarks generation (enabled by default)
# npx repo-to-pdf ../repo --no-outline

-p, --concurrency [num]
set max parallel Puppeteer PDF render jobs (auto uses up to 4 cores)
# npx repo-to-pdf ../repo -p 2

-c, --calibre [path]
path to ebook-convert, for MacOS, try
/Applications/calibre.app/Contents/MacOS/ebook-convert; for linux, try
/usr/bin/ebook-convert
```

For tablet, mobile

only supported by the renderer node

```
npx repo-to-pdf [src/folder] --device tablet
npx repo-to-pdf [src/folder] --device mobile
```

Testing

```
npm run test
```

Performance

```
cd test/data && wget https://github.com/redis/redis/archive/refs/tags/7.0.0.zip  
&& unzip 7.0.0.zip
```

on M1 Macbook Air

```
time npx repo-to-pdf ./test/data/redis-7.0.0/src -s 3  
2.09s user 0.36s system 2% cpu 1:42.14 total  
  
time npx repo-to-pdf ./test/data/redis-7.0.0/src -s 3 -r wkhtmltopdf  
43.78s user 0.84s system 93% cpu 47.787 total
```

Known issues

- The renderer `wkhtmltopdf` can split lines. It's a known unsolved issue of `wkhtmltopdf` :
<https://github.com/wkhtmltopdf/wkhtmltopdf/issues/2141>,
<https://github.com/wkhtmltopdf/wkhtmltopdf/issues/1524>
- To properly install puppeteer on Debian, you might need to install required libs:
<https://github.com/puppeteer/puppeteer/issues/290#issuecomment-322921352> and
`libgbm-dev`

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HEAD

4.3.0 (September 10, 2013)

- Use one apple-touch-icon instead of six ([#1367](#) (<https://github.com/h5bp/html5-boilerplate/issues/1367>)).
- Move font-related declarations from `body` to `html` ([#1411](#) (<https://github.com/h5bp/html5-boilerplate/issues/1411>)).
- Update to Apache Server Configs 1.1.0.
- Add `initial-scale=1` to the viewport `meta` ([#1398](#) (<https://github.com/h5bp/html5-boilerplate/pull/1398>)).
- Vertical centering for audio-, canvas- and video-tags ([#1326](#) (<https://github.com/h5bp/html5-boilerplate/issues/1326>)).
- Remove Google Chrome Frame related code ([#1379](#) (<https://github.com/h5bp/html5-boilerplate/pull/1379>), [#1396](#) (<https://github.com/h5bp/html5-boilerplate/pull/1396>)).

- Update to Google Universal Analytics ([#1347](https://github.com/h5bp/html5-boilerplate/issues/1347) (<https://github.com/h5bp/html5-boilerplate/issues/1347>)).
- Update to jQuery 1.10.2.
- Update to Normalize.css 1.1.3.

4.2.0 (April 8, 2013)

- Remove Google Analytics protocol check ([#1319](https://github.com/h5bp/html5-boilerplate/pull/1319) (<https://github.com/h5bp/html5-boilerplate/pull/1319>)).
- Update to Normalize.css 1.1.1.
- Update Apache configurations to include the latest changes in the canonical `.htaccess` (<https://github.com/h5bp/server-configs-apache>) file.
- Use a protocol relative URL for the 404 template script.
- Update to jQuery 1.9.1.

4.1.0 (January 21, 2013)

- Update to Normalize.css 1.1.0.
- Update to jQuery 1.9.0.

4.0.3 (January 12, 2013)

- Use 32x32 favicon.ico ([#1286](https://github.com/h5bp/html5-boilerplate/pull/1286) (<https://github.com/h5bp/html5-boilerplate/pull/1286>)).
- Remove named function expression in plugins.js ([#1280](https://github.com/h5bp/html5-boilerplate/pull/1280) (<https://github.com/h5bp/html5-boilerplate/pull/1280>)).
- Adjust CSS image-replacement code ([#1239](https://github.com/h5bp/html5-boilerplate/issues/1239) (<https://github.com/h5bp/html5-boilerplate/issues/1239>)).
- Update HiDPI example media query ([#1127](https://github.com/h5bp/html5-boilerplate/issues/1127) (<https://github.com/h5bp/html5-boilerplate/issues/1127>)).

4.0.2 (December 9, 2012)

- Update placeholder icons.
- Update to Normalize.css 1.0.2.
- Update to jQuery 1.8.3.

4.0.1 (October 20, 2012)

- Further improvements to `console` method stubbing ([#1206](https://github.com/h5bp/html5-boilerplate/issues/1206) (<https://github.com/h5bp/html5-boilerplate/issues/1206>), [#1229](https://github.com/h5bp/html5-boilerplate/pull/1229) (<https://github.com/h5bp/html5-boilerplate/pull/1229>)).
- Update to jQuery 1.8.2.
- Update to Modernizr 2.6.2.
- Minor additions to the documentation.

4.0.0 (August 28, 2012)

- Improve the Apache compression configuration (#1012 (<https://github.com/h5bp/html5-boilerplate/issues/1012>), #1173 (<https://github.com/h5bp/html5-boilerplate/issues/1173>)).
- Add a HiDPI example media query (#1127 (<https://github.com/h5bp/html5-boilerplate/issues/1127>)).
- Add bundled docs (#1154 (<https://github.com/h5bp/html5-boilerplate/issues/1154>)).
- Add MIT license (#1139 (<https://github.com/h5bp/html5-boilerplate/issues/1139>)).
- Update to Normalize.css 1.0.1.
- Separate Normalize.css from the rest of the CSS (#1160 (<https://github.com/h5bp/html5-boilerplate/issues/1160>)).
- Improve `console.log` protection (#1107 (<https://github.com/h5bp/html5-boilerplate/issues/1107>)).
- Replace hot pink text selection color with a neutral color.
- Change image replacement technique (#1149 (<https://github.com/h5bp/html5-boilerplate/issues/1149>)).
- Code format and consistency changes (#1112 (<https://github.com/h5bp/html5-boilerplate/issues/1112>)).
- Rename CSS file and rename JS files and subdirectories.
- Update to jQuery 1.8 (#1161 (<https://github.com/h5bp/html5-boilerplate/issues/1161>)).
- Update to Modernizr 2.6.1 (#1086 (<https://github.com/h5bp/html5-boilerplate/issues/1086>)).
- Remove uncompressed jQuery (#1153 (<https://github.com/h5bp/html5-boilerplate/issues/1153>)).
- Remove superfluous inline comments (#1150 (<https://github.com/h5bp/html5-boilerplate/issues/1150>)).

3.0.2 (February 19, 2012)

- Update to Modernizr 2.5.3.

3.0.1 (February 08, 2012).

- Update to Modernizr 2.5.2 (includes html5shiv 3.3).

3.0.0 (February 06, 2012)

- Improvements to `.htaccess`.
- Improve 404 design.
- Simplify JS folder structure.
- Change `html` IE class names changed to target ranges rather than specific versions of IE.

- Update CSS to include latest normalize.css changes and better typographic defaults (#825 (<https://github.com/h5bp/html5-boilerplate/issues/825>)).
- Update to Modernizr 2.5 (includes yepnope 1.5 and html5shiv 3.2).
- Update to jQuery 1.7.1.
- Revert to async snippet for the Google Analytics script.
- Remove the ant build script (#826 (<https://github.com/h5bp/html5-boilerplate/issues/826>)).
- Remove Respond.js (#816 (<https://github.com/h5bp/html5-boilerplate/issues/816>)).
- Remove the `demo/` directory (#808 (<https://github.com/h5bp/html5-boilerplate/issues/808>)).
- Remove the `test/` directory (#808 (<https://github.com/h5bp/html5-boilerplate/issues/808>)).
- Remove Google Chrome Frame script for IE6 users; replace with links to Chrome Frame and options for alternative browsers.
- Remove `initial-scale=1` from the viewport `meta` (#824 (<https://github.com/h5bp/html5-boilerplate/issues/824>)).
- Remove `defer` from all scripts to avoid legacy IE bugs.
- Remove explicit Site Speed tracking for Google Analytics. It's now enabled by default.

2.0.0 (August 10, 2011)

- Change starting CSS to be based on normalize.css instead of reset.css (#500 (<https://github.com/h5bp/html5-boilerplate/issues/500>)).
- Add Respond.js media query polyfill.
- Add Google Chrome Frame script prompt for IE6 users.
- Simplify the `html` conditional comments for modern browsers and add an `oldie` class.
- Update clearfix to use "micro clearfix".
- Add placeholder CSS MQs for mobile-first approach.
- Add `textarea { resize: vertical; }` to only allow vertical resizing.
- Add `img { max-width: 100%; }` to the print styles; prevents images being truncated.
- Add Site Speed tracking for Google Analytics.
- Update to jQuery 1.6.2 (and use minified by default).
- Update to Modernizr 2.0 Complete, Production minified (includes yepnope, html5shiv, and Respond.js).
- Use `Modernizr.load()` to load the Google Analytics script.
- Much faster build process.
- Add build script options for CSSLint, JSLint, JSHint tools.

- Build script now compresses all images in subfolders.
- Build script now versions files by SHA hash.
- Many `.htaccess` improvements including: disable directory browsing, improved support for all versions of Apache, more robust and extensive HTTP compression rules.
- Remove `handheld.css` as it has very poor device support.
- Remove touch-icon `link` elements from the HTML and include improved touch-icon support.
- Remove the cache-busting query paramaters from files references in the HTML.
- Remove IE6 PNGFix.

1.0.0 (March 21, 2011)

- Rewrite build script to make it more customizable and flexible.
- Add a humans.txt.
- Numerous `.htaccess` improvements (including inline documentation).
- Move the alternative server configurations to the H5BP server configs repo.
- Use a protocol-relative url to reference jQuery and prevent mixed content warnings.
- Optimize the Google Analytics snippet.
- Use Eric Meyer's recent CSS reset update and the HTML5 Doctor reset.
- More robust `sub / sup` CSS styles.
- Add keyboard `.focusable` helper class that extends `.visuallyhidden`.
- Print styles no longer print hash or JavaScript links.
- Add a print reset for IE's proprietary filters.
- Remove IE9-specific conditional class on the `html` element.
- Remove margins from lists within `nav` elements.
- Remove YUI profiling.

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Contributing to HTML5 Boilerplate

♥ [HTML5 Boilerplate \(http://html5boilerplate.com\)](http://html5boilerplate.com) and want to get involved?

Thanks! There are plenty of ways you can help!

Please take a moment to review this document in order to make the contribution process easy and effective for everyone involved.

Following these guidelines helps to communicate that you respect the time of the developers managing and developing this open source project. In return, they should reciprocate that respect in addressing your issue or assessing patches and features.

Using the issue tracker

The [issue tracker](https://github.com/h5bp/html5-boilerplate/issues) (<https://github.com/h5bp/html5-boilerplate/issues>) is the preferred channel for [bug reports](#), [features requests](#) and [submitting pull requests](#), but please respect the following restrictions:

- Please **do not** use the issue tracker for personal support requests (use [Stack Overflow](http://stackoverflow.com/questions/tagged/html5boilerplate) (<http://stackoverflow.com/questions/tagged/html5boilerplate>) or IRC).
- Please **do not** derail or troll issues. Keep the discussion on topic and respect the opinions of others.
- Please **do not** open issues or pull requests regarding the code in `.htaccess` (<https://github.com/h5bp/server-configs-apache>), `jQuery` (<https://github.com/jquery/jquery/>), `Modernizr` (<https://github.com/Modernizr/Modernizr>) or `Normalize.css` (<https://github.com/necolas/normalize.css>) (open them in their respective repositories).

Bug reports

A bug is a *demonstrable problem* that is caused by the code in the repository. Good bug reports are extremely helpful - thank you!

Guidelines for bug reports:

1. **Use the GitHub issue search** — check if the issue has already been reported.
2. **Check if the issue has been fixed** — try to reproduce it using the latest `master` or development branch in the repository.
3. **Isolate the problem** — ideally create a [reduced test case](http://css-tricks.com/6263-reduced-test-cases/) (<http://css-tricks.com/6263-reduced-test-cases/>) and a live example.

A good bug report shouldn't leave others needing to chase you up for more information. Please try to be as detailed as possible in your report. What is your environment? What steps will reproduce the issue? What browser(s) and OS experience the problem? What would you expect to be the outcome? All these details will help people to fix any potential bugs.

Example:

Short and descriptive example bug report title

A summary of the issue and the browser/OS environment in which it occurs. If suitable, include the steps required to reproduce the bug.

1. This is the first step
2. This is the second step
3. Further steps, etc.

`<url>` - a link to the reduced test case

Any other information you want to share that is relevant to the issue being reported. This might include the lines of code that you have identified as causing the bug, and potential solutions (and your opinions on their merits).

`<a name="features"></a>`

Feature requests

Feature requests are welcome. But take a moment to find out whether your idea fits with the scope and aims of the project. It's up to *you* to make a strong case to convince the project's developers of the merits of this feature. Please provide as much detail and context as possible.

`<a name="pull-requests"></a>`

Pull requests

Good pull requests - patches, improvements, new features - are a fantastic help. They should remain focused in scope and avoid containing unrelated commits.

Please ask first before embarking on any significant pull request (e.g. implementing features, refactoring code, porting to a different language), otherwise you risk spending a lot of time working on something that the project's developers might not want to merge into the project.

Please adhere to the coding conventions used throughout a project (indentation, accurate comments, etc.) and any other requirements (such as test coverage).

Adhering to the following this process is the best way to get your work included in the project:

1. [Fork \(http://help.github.com/fork-a-repo/\)](http://help.github.com/fork-a-repo/) the project, clone your fork, and configure the remotes:

```
# Clone your fork of the repo into the current directory
git clone https://github.com/<your-username>/html5-boilerplate.git
# Navigate to the newly cloned directory
cd html5-boilerplate
# Assign the original repo to a remote called "upstream"
git remote add upstream https://github.com/h5bp/html5-boilerplate.git
```

2. If you cloned a while ago, get the latest changes from upstream:

```
git checkout master
git pull upstream master
```

3. Create a new topic branch (off the main project development branch) to contain your feature, change, or fix:

```
git checkout -b <topic-branch-name>
```

4. Commit your changes in logical chunks. Please adhere to these [git commit message guidelines \(http://tbaggery.com/2008/04/19/a-note-about-git-commit-messages.html\)](http://tbaggery.com/2008/04/19/a-note-about-git-commit-messages.html)

or your code is unlikely be merged into the main project. Use Git's [interactive rebase \(https://help.github.com/articles/interactive-rebase\)](https://help.github.com/articles/interactive-rebase) feature to tidy up your commits before making them public.

5. Locally merge (or rebase) the upstream development branch into your topic branch:

```
git pull [--rebase] upstream master
```

6. Push your topic branch up to your fork:

```
git push origin <topic-branch-name>
```

7. [Open a Pull Request \(https://help.github.com/articles/using-pull-requests/\)](https://help.github.com/articles/using-pull-requests/) with a clear title and description.

IMPORTANT: By submitting a patch, you agree to allow the project owners to license your work under the the terms of the [MIT License \(LICENSE.md\)](#).

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HTML5 Boilerplate (<http://html5boilerplate.com>)

HTML5 Boilerplate is a professional front-end template for building fast, robust, and adaptable web apps or sites.

This project is the product of many years of iterative development and combined community knowledge. It does not impose a specific development philosophy or framework, so you're free to architect your code in the way that you want.

- Source: <https://github.com/h5bp/html5-boilerplate> (<https://github.com/h5bp/html5-boilerplate>)

- Homepage: <http://html5boilerplate.com> (<http://html5boilerplate.com>)
- Twitter: [@h5bp](http://twitter.com/h5bp) (<http://twitter.com/h5bp>)

Quick start

Choose one of the following options:

1. Download the latest stable release from html5boilerplate.com (<http://html5boilerplate.com/>) or a custom build from [Initializr](http://www.initializr.com) (<http://www.initializr.com>).
2. Clone the git repo — `git clone https://github.com/h5bp/html5-boilerplate.git` - and checkout the tagged release you'd like to use.

Features

- HTML5 ready. Use the new elements with confidence.
- Cross-browser compatible (Chrome, Opera, Safari, Firefox 3.6+, IE6+).
- Designed with progressive enhancement in mind.
- Includes [Normalize.css](http://necolas.github.com/normalize.css/) (<http://necolas.github.com/normalize.css/>) for CSS normalizations and common bug fixes.
- The latest [jQuery](http://jquery.com/) (<http://jquery.com/>) via CDN, with a local fallback.
- The latest [Modernizr](http://modernizr.com/) (<http://modernizr.com/>) build for feature detection.
- IE-specific classes for easier cross-browser control.
- Placeholder CSS Media Queries.
- Useful CSS helpers.
- Default print CSS, performance optimized.
- Protection against any stray `console.log` causing JavaScript errors in IE6/7.
- An optimized Google Analytics snippet.
- Apache server caching, compression, and other configuration defaults for Grade-A performance.
- Cross-domain Ajax and Flash.
- "Delete-key friendly." Easy to strip out parts you don't need.
- Extensive inline and accompanying documentation.

Documentation

Take a look at the [documentation table of contents \(doc/TOC.md\)](#). This documentation is bundled with the project, which makes it readily available for offline reading and provides a useful starting point for any documentation you

want to write about your project.

Contributing

Anyone and everyone is welcome to [contribute \(CONTRIBUTING.md\)](#). Hundreds of developers have helped make the HTML5 Boilerplate what it is today.

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[HTML5 Boilerplate homepage \(http://html5boilerplate.com\)](http://html5boilerplate.com)

HTML5 Boilerplate documentation:

Getting started

- [Usage \(usage.md\)](#) — Overview of the project contents.
- [FAQ \(faq.md\)](#) — Frequently asked questions, along with their answers.

The core of HTML5 Boilerplate

- [HTML \(html.md\)](#) — A guide to the default HTML.
- [CSS \(css.md\)](#) — A guide to the default CSS.
- [JavaScript \(js.md\)](#) — A guide to the default JavaScript.
- [.htaccess \(https://github.com/h5bp/server-configs-apache/tree/master/doc\)](https://github.com/h5bp/server-configs-apache/tree/master/doc) — All about the Apache web server configs (also see our [alternative server configs \(https://github.com/h5bp/server-configs/blob/master/README.md\)](https://github.com/h5bp/server-configs/blob/master/README.md)).
- [crossdomain.xml \(crossdomain.md\)](#) — An introduction to making use of crossdomain requests.
- [Everything else \(misc.md\)](#).

Development

- [Extending and customizing HTML5 Boilerplate \(extend.md\)](#) — Going further with the boilerplate.

Related projects

HTML5 Boilerplate has several related projects to help improve the performance of your site/app in various production environments.

- [Server configs \(https://github.com/h5bp/server-configs\)](https://github.com/h5bp/server-configs) — Configs for different servers.
- [Node build script \(https://github.com/h5bp/node-build-script\)](https://github.com/h5bp/node-build-script) — A feature-rich [grunt \(https://github.com/gruntjs/grunt\)](https://github.com/gruntjs/grunt) plugin.

- [Ant build script \(https://github.com/h5bp/ant-build-script\)](https://github.com/h5bp/ant-build-script) — The original HTML5 Boilerplate build script.

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crossdomain.xml

A cross-domain policy file is an XML document that grants a web client—such as Adobe Flash Player, Adobe Reader, etc., permission to handle data across multiple domains. When a client hosts content from a particular source domain and that content makes requests directed towards a domain other than its own, the remote domain would need to host a cross-domain policy file that grants access to the source domain, allowing the client to continue with the transaction. Policy files grant read access to data, permit a client to include custom headers in cross-domain requests, and are also used with sockets to grant permissions for socket-based connections.

For full details, check out Adobe's article about the [cross-domain policy file specification \(http://www.adobe.com/devnet/articles/crossdomain_policy_file_spec.html\)](http://www.adobe.com/devnet/articles/crossdomain_policy_file_spec.html).

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The CSS

The HTML5 Boilerplate starting CSS includes:

- [Normalize.css \(https://github.com/necolas/normalize.css\)](https://github.com/necolas/normalize.css).
- Useful HTML5 Boilerplate defaults.
- Common helpers.
- Placeholder media queries.
- Print styles.

This starting CSS does not rely on the presence of conditional classnames, conditional style sheets, or Modernizr. It is ready to use whatever your development preferences happen to be.

Normalize.css

Normalize.css is a modern, HTML5-ready alternative to CSS resets. It contains extensive inline documentation. Please refer to the [Normalize.css project \(http://necolas.github.com/normalize.css/\)](http://necolas.github.com/normalize.css/) for more information.

HTML5 Boilerplate defaults

This project includes a handful of base styles that build upon Normalize.css. These include:

- Basic typography settings to provide improved text readability by default.
- Protection against unwanted `text-shadow` during text highlighting.
- Tweaks to default image alignment, fieldsets, and textareas.
- A pretty Chrome Frame prompt.

You are free to modify or add to these base styles as your project requires.

Common helpers

`.ir`

Add the `.ir` class to any element you are applying image-replacement to. When replacing an element's content with an image, make sure to also set a specific `background-image: url(pathtoimage.png);`, `width`, and `height` so that your replacement image appears.

`.hidden`

Add the `.hidden` class to any elements that you want to hide from all presentations, including screen readers. It could be an element that will be populated later with JavaScript or an element you will hide with JavaScript. Do not use this for SEO keyword stuffing. That is just not cool.

`.visuallyhidden`

Add the `.visuallyhidden` class to hide text from browsers but make it available for screen readers. You can use this to hide text that is specific to screen readers but that other users should not see. [About invisible content \(http://www.webaim.org/techniques/css/invisiblecontent/\)](http://www.webaim.org/techniques/css/invisiblecontent/), [Hiding content for accessibility \(http://snook.ca/archives/html_and_css/hiding-content-for-accessibility\)](http://snook.ca/archives/html_and_css/hiding-content-for-accessibility), [HTML5 Boilerplate issue/research \(https://github.com/h5bp/html5-boilerplate/issues/194/\)](https://github.com/h5bp/html5-boilerplate/issues/194/).

`.invisible`

Add the `.invisible` class to any element you want to hide without affecting layout. When you use `display: none` an element is effectively removed from the layout. But in some cases you want the element to simply be invisible while remaining in the flow and not affecting the positioning of surrounding content.

`.clearfix`

Adding `.clearfix` to an element will ensure that it always fully contains its floated children. There have been many variants of the clearfix hack over the years, and there are other hacks that can also help you to contain floated children, but the HTML5 Boilerplate currently uses the [micro clearfix](http://nicolasgallagher.com/micro-clearfix-hack/) (<http://nicolasgallagher.com/micro-clearfix-hack/>).

Media Queries

The boilerplate makes it easy to get started with a "Mobile First" and [Responsive Web Design](http://www.alistapart.com/articles/responsive-web-design/) (<http://www.alistapart.com/articles/responsive-web-design/>) approach to development. But it's worth remembering that there are [no silver bullets](http://www.cloudfour.com/css-media-query-for-mobile-is-fools-gold/) (<http://www.cloudfour.com/css-media-query-for-mobile-is-fools-gold/>).

We include a placeholder Media Queries to build up your mobile styles for wider viewports and high-resolution displays. It's recommended that you adapt these Media Queries based on the content of your site rather than mirroring the fixed dimensions of specific devices.

If you do not want to take a "Mobile First" approach, you can simply edit or remove these placeholder Media Queries. One possibility would be to work from wide viewports down and use `max-width` MQs instead, e.g., `@media only screen and (max-width: 480px)`.

Take a look into the [Mobile Boilerplate](https://github.com/h5bp/mobile-boilerplate) (<https://github.com/h5bp/mobile-boilerplate>) for features that are useful when developing mobile web apps.

Print styles

- Print styles are inlined to [reduce the number of page requests](http://www.phpied.com/delay-loading-your-print-css/) (<http://www.phpied.com/delay-loading-your-print-css/>).
- We strip all background colors, change the font color to black and remove text-shadow. This is meant to [help save printer ink and make the printing process much faster](http://www.sanbeiji.com/archives/953) (<http://www.sanbeiji.com/archives/953>).
- Anchors do not need colors to indicate they are linked. They are underlined to indicate so.

- Anchors and Abbreviations are expanded to indicate where users reading the printed page can refer to.
- But we do not want to show link text for image replaced elements (given that they are primarily images).

Paged media styles

- Paged media is supported only in a [few browsers](#) (http://en.wikipedia.org/wiki/Comparison_of_layout_engines_%28Cascading_Style_Sheets%29#Grammar_and_rules).
- Paged media support means browsers would know how to interpret instructions on breaking content into pages and on orphans/widows.
- We use `page-break-inside: avoid;` to prevent an image and table row from being split into two different pages, so use the same `page-break-inside: avoid;` for that as well.
- Headings should always appear with the text they are titles for. So, we ensure headings never appear in a different page than the text they describe by using `page-break-after: avoid;`.
- We also apply a default margin for the page specified in `cm`.
- We do not want [orphans and widows](#) (http://en.wikipedia.org/wiki/Widows_and_orphans) to appear on pages you print. So, by defining `orphans: 3` and `widows: 3` you define the minimal number of words that every line should contain.

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Extend and customise HTML5 Boilerplate

Here is some useful advice for how you can make your project with HTML5 Boilerplate even better. We don't want to include it all by default, as not everything fits with everyone's needs.

DNS prefetching

In short, DNS Prefetching is a method of informing the browser of domain names referenced on a site so that the client can resolve the DNS for those hosts, cache them, and when it comes time to use them, have a faster turn around on the request.

Implicit prefetches

There is a lot of prefetching done for you automatically by the browser. When the browser encounters an anchor in your html that does not share the same domain name as the current location the browser requests, from the client OS, the IP address for this new domain. The client first checks its cache and then, lacking a cached copy, makes a request from a DNS server. These requests happen in the background and are not meant to block the rendering of the page.

The goal of this is that when the foreign IP address is finally needed it will already be in the client cache and will not block the loading of the foreign content. Less requests result in faster page load times. The perception of this is increased on a mobile platform where DNS latency can be greater.

Disable implicit prefetching

```
<meta http-equiv="x-dns-prefetch-control" content="off">
```

Even with X-DNS-Prefetch-Control meta tag (or http header) browsers will still prefetch any explicit dns-prefetch links.

WARNING: THIS MAY MAKE YOUR SITE SLOWER IF YOU RELY ON RESOURCES FROM FOREIGN DOMAINS.

Explicit prefetches

Typically the browser only scans the HTML for foreign domains. If you have resources that are outside of your HTML (a javascript request to a remote server or a CDN that hosts content that may not be present on every page of your site, for example) then you can queue up a domain name to be prefetched.

```
<link rel="dns-prefetch" href="//example.com">
<link rel="dns-prefetch" href="//ajax.googleapis.com">
```

You can use as many of these as you need, but it's best if they are all immediately after the [Meta Charset](https://developer.mozilla.org/en/HTML/Element/meta#attr-charset) (<https://developer.mozilla.org/en/HTML/Element/meta#attr-charset>) element (which should go right at the top of the `head`), so the browser can act on them ASAP.

Common Prefetch Links

Amazon S3:

```
<link rel="dns-prefetch" href="//s3.amazonaws.com">
```

Google APIs:

```
<link rel="dns-prefetch" href="//ajax.googleapis.com">
```

Microsoft Ajax Content Delivery Network:

```
<link rel="dns-prefetch" href="//ajax.microsoft.com">  
<link rel="dns-prefetch" href="//ajax.aspnetcdn.com">
```

Browser support for DNS prefetching

Chrome, Firefox 3.5+, Safari 5+, Opera (Unknown), IE 9 (called "Pre-resolution" on blogs.msdn.com)

Further reading about DNS prefetching

- https://developer.mozilla.org/En/Controlling_DNS_prefetching
- <http://dev.chromium.org/developers/design-documents/dns-prefetching>
- <http://www.apple.com/safari/whats-new.html>
- <http://blogs.msdn.com/b/ie/archive/2011/03/17/internet-explorer-9-network-performance-improvements.aspx>
- http://dayofjs.com/videos/22158462/web-browsers_alex-russel

Search

Direct search spiders to your sitemap

Learn how to make a sitemap (<http://www.sitemaps.org/protocol.php>)

```
<link rel="sitemap" type="application/xml" title="Sitemap" href="/sitemap.xml">
```

Hide pages from search engines

According to Heather Champ, former community manager at Flickr, you should not allow search engines to index your "Contact Us" or "Complaints" page if you value your sanity. This is an HTML-centric way of achieving that.

```
<meta name="robots" content="noindex">
```

WARNING: DO NOT INCLUDE ON PAGES THAT SHOULD APPEAR IN SEARCH ENGINES.

Firefox and IE Search Plugins

Sites with in-site search functionality should be strongly considered for a browser search plugin. A "search plugin" is an XML file which defines how your plugin behaves in the browser. [How to make a browser search plugin \(http://www.google.com/search?ie=UTF-8&q=how+to+make+browser+search+plugin\)](http://www.google.com/search?ie=UTF-8&q=how+to+make+browser+search+plugin).

```
<link rel="search" title="" type="application/opensearchdescription+xml" href="">
```

Internet Explorer

Prompt users to switch to "Desktop Mode" in IE10 Metro

IE10 does not support plugins, such as Flash, in Metro mode. If your site requires plugins, you can let users know that via the X-UA-Compatible meta element, which will prompt them to switch to Desktop Mode.

```
<meta http-equiv="X-UA-Compatible" content="requiresActiveX=true">
```

Here's what it looks like alongside H5BP's default X-UA-Compatible values:

```
<meta http-equiv="X-UA-Compatible" content="IE=edge,chrome=1,requiresActiveX=true">
```

You can find more information in [Microsoft's IEBlog post about prompting for plugin use in IE10 Metro Mode \(http://blogs.msdn.com/b/ie/archive/2012/01/31/web-sites-and-a-plug-in-free-web.aspx\)](http://blogs.msdn.com/b/ie/archive/2012/01/31/web-sites-and-a-plug-in-free-web.aspx).

IE Pinned Sites (IE9+)

Enabling your application for pinning will allow IE9 users to add it to their Windows Taskbar and Start Menu. This comes with a range of new tools that you can easily configure with the elements below. See more [documentation on IE9 Pinned Sites \(http://msdn.microsoft.com/en-us/library/gg131029.aspx\)](http://msdn.microsoft.com/en-us/library/gg131029.aspx).

Name the Pinned Site for Windows

Without this rule, Windows will use the page title as the name for your application.

```
<meta name="application-name" content="Sample Title">
```

Give your Pinned Site a tooltip

You know — a tooltip. A little textbox that appears when the user holds their mouse over your Pinned Site's icon.

```
<meta name="msapplication-tooltip" content="A description of what this site does.">
```

Set a default page for your Pinned Site

If the site should go to a specific URL when it is pinned (such as the homepage), enter it here. One idea is to send it to a special URL so you can track the number of pinned users, like so:

```
http://www.example.com/index.html?pinned=true
```

```
<meta name="msapplication-starturl" content="http://www.example.com/index.html?pinned=true">
```

Recolor IE's controls manually for a Pinned Site

IE9+ will automatically use the overall color of your Pinned Site's favicon to shade its browser buttons. UNLESS you give it another color here. Only use named colors (`red`) or hex colors (`#ff0000`).

```
<meta name="msapplication-navbutton-color" content="#ff0000">
```

Manually set the window size of a Pinned Site

If the site should open at a certain window size once pinned, you can specify the dimensions here. It only supports static pixel dimensions. 800x600 minimum.

```
<meta name="msapplication-window" content="width=800;height=600">
```

Jump List "Tasks" for Pinned Sites

Add Jump List Tasks that will appear when the Pinned Site's icon gets a right-click. Each Task goes to the specified URL, and gets its own mini icon (essentially a favicon, a 16x16 .ICO). You can add as many of these as you need.

```
<meta name="msapplication-task" content="name=Task 1;action-uri=http://host/Page1.html;icon-uri=http://host/icon1.ico">
<meta name="msapplication-task" content="name=Task 2;action-uri=http://microsoft.com/Page2.html;icon-uri=http://host/icon2.ico">
```

(Windows 8) High quality visuals for Pinned Sites

Windows 8 adds the ability for you to provide a PNG tile image and specify the tile's background color. [Full details on the IE](#)

[blog \(http://blogs.msdn.com/b/ie/archive/2012/06/08/high-quality-visuals-for-pinned-sites-in-windows-8.aspx\)](http://blogs.msdn.com/b/ie/archive/2012/06/08/high-quality-visuals-for-pinned-sites-in-windows-8.aspx).

- Create a 144x144 image of your site icon, filling all of the canvas, and using a transparent background.
- Save this image as a 32-bit PNG and optimize it without reducing colour-depth. It can be named whatever you want (e.g. `metro-tile.png`).
- To reference the tile and its color, add the HTML `meta` elements described in the IE Blog post.

(Windows 8) Badges for Pinned Sites

IE10 will poll an XML document for badge information to display on your app's tile in the Start screen. The user will be able to receive these badge updates even when your app isn't actively running. The badge's value can be a number, or one of a predefined list of glyphs.

- [Tutorial on IEBlog with link to badge XML schema \(http://blogs.msdn.com/b/ie/archive/2012/04/03/pinned-sites-in-windows-8.aspx\)](http://blogs.msdn.com/b/ie/archive/2012/04/03/pinned-sites-in-windows-8.aspx)
- [Available badge values \(http://msdn.microsoft.com/en-us/library/ie/br212849.aspx\)](http://msdn.microsoft.com/en-us/library/ie/br212849.aspx)

```
<meta name="msapplication-badge" value="frequency=NUMBER_IN_MINUTES;polling-uri=http://www.example.com/path/to/file.xml">
```

Disable link highlighting upon tap in IE10

Similar to `-webkit-tap-highlight-color` (<http://davidwalsh.name/mobile-highlight-color>) in iOS Safari. Unlike that CSS property, this is an HTML meta element, and it's value is boolean rather than a color. It's all or nothing.

```
<meta name="msapplication-tap-highlight" content="no" />
```


You can read about this useful element and more techniques in [Microsoft's documentation on adapting WebKit-oriented apps for IE10](http://blogs.windows.com/windows_phone/b/wpdev/archive/2012/11/15/adapting-your-webkit-optimized-site-for-internet-explorer-10.aspx) (http://blogs.windows.com/windows_phone/b/wpdev/archive/2012/11/15/adapting-your-webkit-optimized-site-for-internet-explorer-10.aspx).

Suppress IE6 image toolbar

Kill IE6's pop-up-on-mouseover toolbar for images that can interfere with certain designs and be pretty distracting in general.

```
<meta http-equiv="imagetoolbar" content="false">
```

Social Networks

Facebook Open Graph data

You can control the information that Facebook and others display when users share your site. Below are just the most basic data points you might need. For specific content types (including "website"), see [Facebook's built-in Open Graph content](#)

[templates](https://developers.facebook.com/docs/opengraph/objects/builtin/) (<https://developers.facebook.com/docs/opengraph/objects/builtin/>).

Take full advantage of Facebook's support for complex data and activity by following the [Open Graph](#)

[tutorial](https://developers.facebook.com/docs/opengraph/tutorial/) (<https://developers.facebook.com/docs/opengraph/tutorial/>).

```
<meta property="og:title" content="">
<meta property="og:description" content="">
<meta property="og:image" content="">
```

Twitter Cards

Twitter provides a snippet specification that serves a similar purpose to Open Graph. In fact, Twitter will use Open Graph when Cards is not available. Note that, as of this writing, Twitter requires that app developers activate Cards on a per-domain basis. You can read more about the various snippet formats and application process in the [official Twitter Cards documentation](#) (<https://dev.twitter.com/docs/cards>).

```
<meta name="twitter:card" content="summary">
<meta name="twitter:site" content="@site_account">
<meta name="twitter:creator" content="@individual_account">
<meta name="twitter:url" content="http://www.example.com/path/to/page.html">
<meta name="twitter:title" content="">
<meta name="twitter:description" content="">
<meta name="twitter:image" content="http://www.example.com/path/to/image.jpg">
```

URLs

Canonical URL

Signal to search engines and others "Use this URL for this page!" Useful when parameters after a `#` or `?` is used to control the display state of a page.

`http://www.example.com/cart.html?shopping-cart-open=true` can be indexed as the cleaner, more accurate `http://www.example.com/cart.html`.

```
<link rel="canonical" href="">
```

Official shortlink

Signal to the world "This is the shortened URL to use this page!" Poorly supported at this time. Learn more by reading the [article about shortlinks on the Microformats wiki](http://microformats.org/wiki/rel-shortlink) (<http://microformats.org/wiki/rel-shortlink>).

```
<link rel="shortlink" href="h5bp.com">
```

News Feeds

RSS

Have an RSS feed? Link to it here. Want to [learn how to write an RSS feed from scratch](http://www.rssboard.org/rss-specification) (<http://www.rssboard.org/rss-specification>)?

```
<link rel="alternate" type="application/rss+xml" title="RSS" href="/rss.xml">
```

Atom

Atom is similar to RSS, and you might prefer to use it instead of or in addition to it. [See what Atom's all about](http://www.atomenabled.org/developers/syndication/) (<http://www.atomenabled.org/developers/syndication/>).

```
<link rel="alternate" type="application/atom+xml" title="Atom" href="/atom.xml">
```

Pingbacks

Your server may be notified when another site links to yours. The href attribute should contain the location of your pingback service.

```
<link rel="pingback" href="">
```

- High-level explanation:
http://codex.wordpress.org/Introduction_to_Blogging#Pingbacks
- Step-by-step example case: <http://www.hixie.ch/specs/pingback/pingback-1.0#TOC5>
- PHP pingback service: <http://blog.perplexedlabs.com/2009/07/15/xmlrpc-pingbacks-using-php/>

App Stores

Install a Chrome Web Store app

Users can install a Chrome app directly from your website, as long as the app and site have been associated via Google's Webmaster Tools. Read more on [Chrome Web Store's Inline Installation docs](https://developers.google.com/chrome/web-store/docs/inline_installation) (https://developers.google.com/chrome/web-store/docs/inline_installation).

```
<link rel="chrome-webstore-item"
href="https://chrome.google.com/webstore/detail/APP_ID">
```

Smart App Banners in iOS 6 Safari

Stop bothering everyone with gross modals advertising your entry in the App Store. This bit of code will unintrusively allow the user the option to download your iOS app, or open it with some data about the user's current state on the website.

```
<meta name="apple-itunes-app" content="app-id=APP_ID,app-argument=SOME_TEXT">
```

Google Analytics augments

More tracking settings

The [optimized Google Analytics snippet](http://mathiasbynens.be/notes/async-analytics-snippet) (<http://mathiasbynens.be/notes/async-analytics-snippet>) included with HTML5 Boilerplate includes something like this:

```
var _gaq = [['_setAccount', 'UA-XXXXX-X'], ['_trackPageview']];
```

In case you need more settings, just extend the array literal instead of

`.push()` ing to the

array (<http://mathiasbynens.be/notes/async-analytics-snippet#dont-push-it>)

afterwards:

```
var _gaq = [['_setAccount', 'UA-XXXXX-X'],['_trackPageview'],
['_setAllowAnchor', true]];
```

Anonymize IP addresses

In some countries, no personal data may be transferred outside jurisdictions that do not have similarly strict laws (i.e. from Germany to outside the EU).

Thus a webmaster using the Google Analytics script may have to ensure that no personal (trackable) data is transferred to the US. You can do that with the

`_gat.anonymizeIp`

option

(http://code.google.com/apis/analytics/docs/gaJS/gaJSApi_gat.html#_gat._anonymizeIp).

In use it looks like this:

```
var _gaq = [['_setAccount', 'UA-XXXXX-X'],['_gat._anonymizeIp'],
['_trackPageview']];
```

Track jQuery AJAX requests in Google Analytics

An article by @JangoSteve explains how to [track jQuery AJAX requests in Google](http://www.alfajango.com/blog/track-jquery-ajax-requests-in-google-analytics/)

[Analytics \(http://www.alfajango.com/blog/track-jquery-ajax-requests-in-google-analytics/\)](http://www.alfajango.com/blog/track-jquery-ajax-requests-in-google-analytics/).

Add this to `plugins.js` :

```
/*
 * Log all jQuery AJAX requests to Google Analytics
 * See: http://www.alfajango.com/blog/track-jquery-ajax-requests-in-google-
analytics/
 */
if (typeof _gaq !== "undefined" && _gaq !== null) {
    $(document).ajaxSend(function(event, xhr, settings){
        _gaq.push(['_trackPageview', settings.url]);
    });
}
```

Track JavaScript errors in Google Analytics

Add this function after `_gaq` is defined:

```

(function(window){
    var undefined,
    link = function (href) {
        var a = window.document.createElement('a');
        a.href = href;
        return a;
    };
    window.onerror = function (message, file, line, column) {
        var host = link(file).hostname;
        _gaq.push([
            '_trackEvent',
            (host == window.location.hostname || host == undefined || host == ''
? '' : 'external ') + 'error',
            message, file + ' LINE: ' + line + (column ? ' COLUMN: ' + column :
''), undefined, undefined, true
        ]);
    };
})(window));

```

Track page scroll

Add this function after `_gaq` is defined:

```

$(function(){
    var isDuplicateScrollEvent,
    scrollTimeStart = new Date,
    $window = $(window),
    $document = $(document),
    scrollPercent;

    $window.scroll(function() {
        scrollPercent = Math.round(100 * ($window.height() +
$window.scrollTop())/ $document.height());
        if (scrollPercent > 90 && !isDuplicateScrollEvent) { //page scrolled to
90%
            isDuplicateScrollEvent = 1;
            _gaq.push(['_trackEvent', 'scroll',
                'Window: ' + $window.height() + 'px; Document: ' +
$document.height() + 'px; Time: ' + Math.round((new Date - scrollTimeStart
)/1000,1) + 's',
                undefined, undefined, true
            ]);
        }
    });
});

```

iOS Web Apps

There are a couple of meta tags that provide information about a web app when added to the Home Screen on iOS.

Adding `apple-mobile-web-app-capable` will make your web app chrome-less and provide the default iOS app view. You can control the color scheme of the default view by adding `apple-mobile-web-app-status-bar-style`.

```
<meta name="apple-mobile-web-app-capable" content="yes">
<meta name="apple-mobile-web-app-status-bar-style" content="black">
```

You can use `apple-mobile-web-app-title` to add a specific sites name for the Home Screen icon. This works since iOS 6.

```
<meta name="apple-mobile-web-app-title" content="">
```

For further information please read the [official documentation](http://developer.apple.com/library/safari/#documentation/AppleApplications/Reference/SafariHTMLRef/Articles/MetaTags.html) (<http://developer.apple.com/library/safari/#documentation/AppleApplications/Reference/SafariHTMLRef/Articles/MetaTags.html>) on Apple's site.

Apple Touch Icons

Touch Icons can be seen as the favicons of mobile devices and tablets.

If your site or icons are in a sub-directory, you will need to reference the icons using `link` elements placed in the HTML `head` of your document.

```
<link rel="apple-touch-icon-precomposed" href="apple-touch-icon-precomposed.png">
```

The main sizes of the icons on iOS are:

- iPad, high-resolution display, iOS 7: 152x152
- iPad, high-resolution display, iOS ≤ 6: 144x144
- iPhone, high-resolution display, iOS 7: 120x120
- iPhone, high-resolution display, iOS ≤ 6: 114x114
- iPad, non-Retina, iOS ≤ 6: 72x72

For non-Retina iPhone, iPod Touch, and Android 2.1+ devices you can use the example from above or replace the `apple-touch-icon-precomposed.png` within this project's root folder.

Please refer to Mathias' [article on Touch](#)

[Icons \(http://mathiasbynens.be/notes/touch-icons\)](http://mathiasbynens.be/notes/touch-icons) for a comprehensive overview.

Apple Touch Startup Image

Apart from that it is possible to add start-up screens for web apps on iOS. This basically works by defining `apple-touch-startup-image` with an according link to the image. Since iOS devices have different screen resolutions it is necessary to add media queries to detect which image to load. Here is an example for a retina iPhone:

```
<link rel="apple-touch-startup-image" media="(max-device-width: 480px) and (-webkit-min-device-pixel-ratio: 2)" href="img/startup-retina.png">
```

However, it is possible to detect which start-up image to use with JavaScript.

The Mobile Boilerplate provides a useful function for this. Please see

[helpers.js \(https://github.com/h5bp/mobile-boilerplate/blob/master/js/helper.js#L354\)](https://github.com/h5bp/mobile-boilerplate/blob/master/js/helper.js#L354) for the implementation.

Miscellaneous

- Use [HTML5 polyfills \(https://github.com/Modernizr/Modernizr/wiki/HTML5-Cross-browser-Polyfills\)](#).
- Use [Microformats \(http://microformats.org/wiki/Main_Page\)](http://microformats.org/wiki/Main_Page) (via [microdata \(http://microformats.org/wiki/microdata\)](http://microformats.org/wiki/microdata)) for optimum search results
[visibility \(http://googlewebmastercentral.blogspot.com/2009/05/introducing-rich-snippets.html\)](http://googlewebmastercentral.blogspot.com/2009/05/introducing-rich-snippets.html).
- If you're building a web app you may want [native style momentum scrolling in iOS5 \(http://johanbrook.com/browsers/native-momentum-scrolling-ios-5/\)](http://johanbrook.com/browsers/native-momentum-scrolling-ios-5/) using `-webkit-overflow-scrolling: touch`.
- Avoid development/stage websites "leaking" into SERPs (search engine results page) by [implementing X-Robots-tag headers \(https://github.com/h5bp/html5-boilerplate/issues/804\)](#).

- Screen readers currently have less-than-stellar support for HTML5 but the JS script [accessifyhtml5.js](https://github.com/yatil/accessifyhtml5.js) (<https://github.com/yatil/accessifyhtml5.js>) can help increase accessibility by adding ARIA roles to HTML5 elements.

Many thanks to [Brian Blakely](https://github.com/brianblakely) (<https://github.com/brianblakely>) for contributing much of this information.

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Frequently asked questions

Why is the URL for jQuery without "http"?

This is an intentional use of [protocol-relative URLs](#) (<http://paulirish.com/2010/the-protocol-relative-url/>)

N.B. Using a protocol-relative URL for files that exist on a CDN is problematic when you try to view your local files directly in the browser. The browser will attempt to fetch the file from your local file system. We recommend that you use a local server to test your pages (or Dropbox). This can be done using Python 2.x by running `python -m SimpleHTTPServer` or Python 3.x with `python -m http.server` from your local directory, using Ruby by installing and running [asdf](https://rubygems.org/gems/asdf) (<https://rubygems.org/gems/asdf>), and by installing any one of XAMPP, MAMP, or WAMP.

Why don't you automatically load the latest version of jQuery from the Google CDN?

1. The latest version of jQuery may not be compatible with the existing plugins/code on the site. Version updating should be an intentional decision.
2. The latest version has a very short `max-age=3600` compares to the specific version of `max-age=31536000`, which means you won't get the benefits of long-term caching.

Why is the Google Analytics code at the bottom? Google recommends it be placed the `head`.

The advantage to placing it in the `head` is that you will track a user's pageview even if they leave the page before it has been fully loaded. However, putting the code at the bottom keeps all the scripts together and reinforces that scripts at the bottom are the right move.

How can I integrate [Twitter Bootstrap](http://twitter.github.com/bootstrap/) (<http://twitter.github.com/bootstrap/>) with HTML5 Boilerplate?

You can use [Initializr](http://initializr.com) (<http://initializr.com>) to create a custom build that includes HTML5 Boilerplate with Twitter Bootstrap.

Read more about how [HTML5 Boilerplate and Twitter Bootstrap complement each other](http://www.quora.com/Is-Bootstrap-a-complement-OR-an-alternative-to-HTML5-Boilerplate-or-viceversa/answer/Nicolas-Gallagher) (<http://www.quora.com/Is-Bootstrap-a-complement-OR-an-alternative-to-HTML5-Boilerplate-or-viceversa/answer/Nicolas-Gallagher>).

How do I prevent phone numbers looking twice as large and having a Skype highlight?

If this is occurring, it is because a user has the Skype browser extension installed.

Use the following CSS to prevent Skype from formatting the numbers on your page:

```
span.skype_pnh_container {  
    display: none !important;  
}  
  
span.skype_pnh_print_container {  
    display: inline !important;  
}
```

Do I need to upgrade my sites each time a new version of HTML5 Boilerplate is released?

No. You don't normally replace the foundations of a house once it has been built. There is nothing stopping you from trying to work in the latest changes but you'll have to assess the costs/benefits of doing so.

Where can I get help for support questions?

Please ask for help on

[StackOverflow](http://stackoverflow.com/questions/tagged/html5boilerplate) (<http://stackoverflow.com/questions/tagged/html5boilerplate>).

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The HTML

Conditional `html` classes

A series of IE conditional comments apply the relevant IE-specific classes to the `html` tag. This provides one method of specifying CSS fixes for specific legacy versions of IE. While you may or may not choose to use this technique in your project code, HTML5 Boilerplate's default CSS does not rely on it.

When using the conditional classes technique, applying classes to the `html` element has several benefits:

- It avoids a [file blocking issue](http://webforscher.wordpress.com/2010/05/20/ie-6-slowness-down-ie-8/) (<http://webforscher.wordpress.com/2010/05/20/ie-6-slowness-down-ie-8/>) discovered by Stoyan Stefanov and Markus Leptien.
- It avoids the need for an empty comment that also fixes the above issue.
- CMSes like WordPress and Drupal use the body class more heavily. This makes integrating there a touch simpler.
- It still validates as HTML5.
- It uses the same element as Modernizr (and Dojo). That feels nice.
- It can improve the clarity of code in multi-developer teams.

The `no-js` class

Allows you to more easily explicitly add custom styles when JavaScript is disabled (`no-js`) or enabled (`js`). More here: [Avoiding the FOUC](http://paulirish.com/2009/avoiding-the-fouc-v3/) (<http://paulirish.com/2009/avoiding-the-fouc-v3/>).

``anchor=true`>The order of meta tags, and `<title>`

As recommended by [the HTML5 spec](http://www.whatwg.org/specs/web-apps/current-work/complete/semantics.html#charset) (<http://www.whatwg.org/specs/web-apps/current-work/complete/semantics.html#charset>) (4.2.5.5 Specifying the document's character encoding), add your charset declaration early (before any ASCII art ;) to avoid a potential [encoding-related security issue](http://code.google.com/p/doctype-mirror/wiki/ArticleUtf7) (<http://code.google.com/p/doctype-mirror/wiki/ArticleUtf7>) in IE. It should come in the first `1024` bytes (<http://www.whatwg.org/specs/web-apps/current-work/multipage/semantics.html#charset>).

The charset should also come before the `<title>` tag, due to [potential XSS vectors](http://code.google.com/p/doctype-mirror/wiki/ArticleUtf7) (<http://code.google.com/p/doctype-mirror/wiki/ArticleUtf7>).

The meta tag for compatibility mode [needs to be before all elements except title and meta \(http://h5bp.com/f\)](http://h5bp.com/f).

And that same meta tag can only be invoked for Google Chrome Frame if it is within the [first 1024](http://code.google.com/p/chromium/issues/detail?id=23003)

bytes (<http://code.google.com/p/chromium/issues/detail?id=23003>).

X-UA-Compatible

This makes sure the latest version of IE is used in versions of IE that contain multiple rendering engines. Even if a site visitor is using IE8 or IE9, it's possible that they're not using the latest rendering engine their browser contains. To fix this, use:

```
<meta http-equiv="X-UA-Compatible" content="IE=edge">
```

The `meta` tag tells the IE rendering engine it should use the latest, or edge, version of the IE rendering environment.

This `meta` tag ensures that anyone browsing your site in IE is treated to the best possible user experience that their browser can offer.

This line breaks validation. To avoid this edge case issue it is recommended that you **remove this line and use the** `.htaccess` (or other server config) to send these headers instead. You also might want to read [Validating:](#)

[X-UA-Compatible](#)

(http://groups.google.com/group/html5boilerplate/browse_thread/thread/6d1b6b152aca8ed2).

If you are serving your site on a non-standard port, you will need to set this header on the server-side. This is because the IE preference option 'Display intranet sites in Compatibility View' is checked by default.

Mobile viewport

There are a few different options that you can use with the `viewport meta tag` (https://docs.google.com/present/view?id=dkx3qtm_22dxsrgcf4). You can find out more in [the](#)

[Apple developer docs \(http://j.mp/mobileviewport\)](http://j.mp/mobileviewport). HTML5 Boilerplate comes with a simple setup that strikes a good balance for general use cases.

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

Favicons and Touch Icon

The shortcut icons should be put in the root directory of your site. HTML5 Boilerplate comes with a default set of icons (include favicon and one Apple Touch Icon) that you can use as a baseline to create your own.

Please refer to the more detailed description in the [Extend section \(extend.md\)](#) of these docs.

Modernizr

HTML5 Boilerplate uses a custom build of Modernizr.

[Modernizr \(http://modernizr.com\)](http://modernizr.com) is a JavaScript library which adds classes to the `html` element based on the results of feature test and which ensures that all browsers can make use of HTML5 elements (as it includes the HTML5 Shiv). This allows you to target parts of your CSS and JavaScript based on the features supported by a browser.

In general, in order to keep page load times to a minimum, it's best to call any JavaScript at the end of the page because if a script is slow to load from an external server it may cause the whole page to hang. That said, the Modernizr script *needs* to run *before* the browser begins rendering the page, so that browsers lacking support for some of the new HTML5 elements are able to handle them properly. Therefore the Modernizr script is the only JavaScript file synchronously loaded at the top of the document.

The content area

The central part of the boilerplate template is pretty much empty. This is intentional, in order to make the boilerplate suitable for both web page and web app development.

Google Chrome Frame

The main content area of the boilerplate includes a prompt to install Chrome Frame (which no longer requires administrative rights) for users of IE 6. If you intended to support IE 6, then you should remove the snippet of code.

Google CDN for jQuery

The Google CDN version of the jQuery JavaScript library is referenced towards the bottom of the page using a protocol-independent path (read more about this in the [FAQ \(faq.md\)](#)). A local fallback of jQuery is included for rare instances when the CDN version might not be available, and to facilitate offline development.

Regardless of which JavaScript library you choose to use, it is well worth the time and effort to look up and reference the Google CDN (Content Delivery Network) version. Your users may already have this version cached in their browsers, and Google's CDN is likely to deliver the asset faster than your server.

Google Analytics Tracking Code

Finally, an optimized version of the latest Google Analytics tracking code is included. Google recommends that this script be placed at the top of the page. Factors to consider: if you place this script at the top of the page, you'll be able to count users who don't fully load the page, and you'll incur the max number of simultaneous connections of the browser.

Further information:

- [Optimizing the asynchronous Google Analytics snippet \(http://mathiasbynens.be/notes/async-analytics-snippet\)](http://mathiasbynens.be/notes/async-analytics-snippet).
- [Tracking Site Activity - Google Analytics \(http://code.google.com/apis/analytics/docs/tracking/asyncTracking.html\)](http://code.google.com/apis/analytics/docs/tracking/asyncTracking.html).

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The JavaScript

Information about the default JavaScript included in the project.

main.js

This file can be used to contain or reference your site/app JavaScript code. For larger projects, you can make use of a JavaScript module loader, like [Require.js \(http://requirejs.org/\)](http://requirejs.org/), to load any other scripts you need to run.

plugins.js

This file can be used to contain all your plugins, such as jQuery plugins and other 3rd party scripts.

One approach is to put jQuery plugins inside of a `(function($){ ... })(jQuery);` closure to make sure they're in the jQuery namespace safety blanket. Read more about [jQuery plugin](#)

vendor

This directory can be used to contain all 3rd party library code.

Minified versions of the latest jQuery and Modernizr libraries are included by default. You may wish to create your own [custom Modernizr build](#) (<http://www.modernizr.com/download/>).

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Miscellaneous

.gitignore

HTML5 Boilerplate includes a basic project-level `.gitignore`. This should primarily be used to avoid certain project-level files and directories from being kept under source control. Different development-environments will benefit from different collections of ignores.

OS-specific and editor-specific files should be ignored using a "global ignore" that applies to all repositories on your system.

For example, add the following to your `~/.gitconfig`, where the `.gitignore` in your HOME directory contains the files and directories you'd like to globally ignore:

```
[core]
  excludesfile = ~/.gitignore
```

- More on global ignores: <http://help.github.com/ignore-files/>
- Comprehensive set of ignores on GitHub: <https://github.com/github/gitignore>

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Usage

Once you have cloned or downloaded HTML5 Boilerplate, creating a site or app usually involves the following:

1. Set up the basic structure of the site.
2. Add some content, style, and functionality.
3. Run your site locally to see how it looks.
4. (Optionally run a build script to automate the optimization of your site - e.g. [ant build script \(https://github.com/h5bp/ant-build-script\)](https://github.com/h5bp/ant-build-script) or [node build script \(https://github.com/h5bp/node-build-script\)](https://github.com/h5bp/node-build-script)).
5. Deploy your site.

Basic structure

A basic HTML5 Boilerplate site initially looks something like this:

```
.
├── css
│   ├── main.css
│   └── normalize.css
├── doc
├── img
├── js
│   ├── main.js
│   ├── plugins.js
│   └── vendor
│       ├── jquery.min.js
│       └── modernizr.min.js
├── .htaccess
├── 404.html
├── apple-touch-icon-precomposed.png
├── index.html
├── humans.txt
├── robots.txt
├── crossdomain.xml
└── favicon.ico
```

What follows is a general overview of each major part and how to use them.

CSS

This directory should contain all your project's CSS files. It includes some initial CSS to help get you started from a solid foundation. [About the CSS \(css.md\)](#).

doc

This directory contains all the HTML5 Boilerplate documentation. You can use it as the location and basis for your own project's documentation.

js

This directory should contain all your project's JS files. Libraries, plugins, and custom code can all be included here. It includes some initial JS to help get you started. [About the JavaScript \(js.md\)](#).

.htaccess

The default web server configs are for Apache. For more information, please refer to the [Apache Server Configs documentation \(https://github.com/h5bp/server-configs-apache/tree/master/doc\)](#).

Host your site on a server other than Apache? You're likely to find the corresponding server configs project listed in our [Server Configs \(https://github.com/h5bp/server-configs/blob/master/README.md\)](#) repository.

404.html

A helpful custom 404 to get you started.

index.html

This is the default HTML skeleton that should form the basis of all pages on your site. If you are using a server-side templating framework, then you will need to integrate this starting HTML with your setup.

Make sure that you update the URLs for the referenced CSS and JavaScript if you modify the directory structure at all.

If you are using Google Analytics, make sure that you edit the corresponding snippet at the bottom to include your analytics ID.

humans.txt

Edit this file to include the team that worked on your site/app, and the technology powering it.

robots.txt

Edit this file to include any pages you need hidden from search engines.

crossdomain.xml

A template for working with cross-domain requests. [About crossdomain.xml \(crossdomain.md\)](#).

Icons

Replace the default `favicon.ico` and Apple Touch Icon with your own.

If you want to use different Apple Touch Icons for different resolutions please refer to the [according documentation \(extend.md#apple-touch-icons\)](#).

You might want to check out Hans' handy [HTML5 Boilerplate Favicon and Apple Touch Icon](#)

[PSD-Template \(http://drublic.de/blog/html5-boilerplate-favicons-psd-template/\)](http://drublic.de/blog/html5-boilerplate-favicons-psd-template/).

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```
// Avoid `console` errors in browsers that lack a console.
(function() {
  var method;
  var noop = function () {};
  var methods = [
    'assert', 'clear', 'count', 'debug', 'dir', 'dirxml', 'error',
    'exception', 'group', 'groupCollapsed', 'groupEnd', 'info', 'log',
    'markTimeline', 'profile', 'profileEnd', 'table', 'time', 'timeEnd',
    'timeStamp', 'trace', 'warn'
  ];
  var length = methods.length;
  var console = (window.console = window.console || {});

  while (length--) {
    method = methods[length];

    // Only stub undefined methods.
    if (!console[method]) {
      console[method] = noop;
    }
  }
}());

// Place any jQuery/helper plugins in here.
```

src/bin/index.js


```

#!/usr/bin/env node
const program = require('commander')

const { getSizeInByte } = require('../utils')
const { generateEbook } = require('../html')

const version = require('../package.json').version

let inputFolder, outputFile
program
  .version('repo-to-pdf ' + version)
  .usage('<input> [output] [options]')
  .arguments('<input> [output] [options]')
  .option('-d, --device <platform>', 'device [desktop(default)|mobile|tablet]',
    'desktop')
  .option('-t, --title [name]', 'title')
  .option('-w, --whitelist [wlist]', 'file format white list, split by ,')
  .option('-s, --size [size]', 'pdf file size limit, in Mb')
  .option('-r, --renderer <engine>', '[node|wkhtmltopdf|calibre] use
node(puppeteer), wkhtmltopdf or calibre to render pdf', 'node')
  .option('-f, --format <ext>', 'output format, pdf|mobi|epub', 'pdf')
  .option('--no-outline', 'disable PDF outlines/bookmarks (default enabled)')
  .option('-p, --concurrency <num>', 'number of parallel Puppeteer PDF render
jobs')
  .option('-c, --calibre [path]', 'path to calibre')
  .option('-b, --baseUrl [url]', 'base url of html folder. By default file:// is
used.')
  .action(function(input, output) {
    inputFolder = input
    outputFile = output
  })

program.parse(process.argv)

const PDF_SIZE      = getSizeInByte(10) // 10 Mb
const title         = program.title || inputFolder
const device        = program.device
const pdf_size      = program.size ? getSizeInByte(program.size) : PDF_SIZE
const format        = program.format
const white_list    = program.whitelist
const renderer      = program.renderer
const calibrePath   = program.calibre
const baseUrl       = program.baseUrl
const outline       = program.outline
const concurrency   = program.concurrency

generateEbook(inputFolder, outputFile, title, {
  renderer,
  calibrePath,
  pdf_size,

```

```
white_list,  
format,  
device,  
baseUrl,  
outline,  
concurrency,  
}).catch((error) => {  
  console.error(error)  
  process.exitCode = 1  
})
```

src/html.js

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```

const path = require('path')
const fs = require('fs')
const os = require('os')

const { Remarkable } = require('remarkable')
const hljs = require('highlight.js')

const RepoBook = require('./repo')
const { sequenceRenderEbook, sequenceRenderPDF, renderPDF } =
require('./render')
const { getFileName, getFileNameExt } = require('./utils')

function getRemarkableParser() {
  return new Remarkable({
    breaks: true,
    highlight: function(str, lang) {
      if (lang && hljs.getLanguage(lang)) {
        try {
          return hljs.highlight(lang, str).value
        } catch (err) {}
      }

      try {
        return hljs.highlightAuto(str).value
      } catch (err) {}

      return ''
    },
  }).use(function(remarkable) {
    remarkable.renderer.rules.heading_open = function(tokens, idx) {
      return '<h' + tokens[idx].hLevel + ' id=' + tokens[idx + 1].content + '
anchor=true>'
    }
  })
}

// => './path/file-1.html'
function getHTMLFiles(mdString, repoBook, options) {
  const { pdf_size, white_list, device, baseUrl, protocol, renderer,
outputFileName, inputFolder } = options
  const opts = {
    cssPath: {
      desktop: '/css/github-min.css',
      tablet: '/css/github-min-tablet.css',
      mobile: '/css/github-min-mobile.css',
    },
    highlightCssPath: '/css/vs.css',
    relaxedCSS: {
      desktop: '',
      tablet: `@page {

```

```

        size: 8in 14in;
        -relaxed-page-width: 8in;
        -relaxed-page-height: 14in;
        margin: 0;
    }`,
    mobile: `@page {
        size: 6in 10in;
        -relaxed-page-width: 6in;
        -relaxed-page-height: 10in;
        margin: 0;
    }`,
},
}
const HTMLFileNameWithExt = getFileNameExt(outputFileName, 'html') ||
getFileName(inputFolder) + '.html'
let outputFile = path.resolve(process.cwd(), HTMLFileNameWithExt)

const mdParser = getRemarkableParser()

const mdHtml = `<article class="markdown-body">` + mdParser.render(mdString) +
`</article>`
const indexHtmlPath = path.join(__dirname, '../html5bp', 'index.html')
const htmlString = fs
    .readFileSync(indexHtmlPath, 'utf-8')
    // TODO: this sits before content replacing, to prevent replacing baseUrl in
content text
    .replace(/\\{\\{baseUrl\\}\\}/g, protocol + baseUrl)
    .replace(`{{cssPath}}`, protocol + baseUrl + opts.cssPath[device])
    .replace(`{{highlightPath}}`, protocol + baseUrl + opts.highlightCssPath)
    .replace(`{{relaxedCSS}}`, opts.relaxedCSS[device])
    .replace(`{{content}}`, mdHtml)

if (!repoBook.hasSingleFile()) {
    outputFile = outputFile.replace('.html', '-' + repoBook.currentPart() +
'.html')
}
fs.writeFileSync(outputFile, htmlString)
return outputFile
}

function checkOptions(options) {
    if (options.outline === undefined) {
        options.outline = true
    }

    if (options.concurrency !== undefined) {
        const concurrency = Number(options.concurrency)
        if (!Number.isInteger(concurrency) || concurrency < 1) {
            console.log(`Invalid concurrency "${options.concurrency}", fallback to
auto mode.`)
            delete options.concurrency
        }
    }
}

```

```

    } else {
        options.concurrency = concurrency
    }
}

if (options.baseUrl) {
    options.protocol = ''
} else {
    options.protocol = os.name === 'windows' ? 'file:///': 'file://'
    options.baseUrl = path.resolve(__dirname, '../html5bp')
}

if (options.format !== 'pdf' && options.renderer === 'node') {
    console.log(`Try to create ${options.format}, use renderer Calibre.`)
    options.renderer = 'calibre'
}

// check calibre path
const calibrePaths = ['/Applications/calibre.app/Contents/MacOS/ebook-convert', '/usr/bin/ebook-convert']
if (options.calibrePath) {
    calibrePaths.unshift(options.calibrePath)
}

let i = 0
for (; i < calibrePaths.length; i++) {
    if (fs.existsSync(calibrePaths[i])) {
        options.calibrePath = calibrePaths[i]
        break
    }
}
if (i === calibrePaths.length && ['mobi', 'epub'].includes(options.format)) {
    console.log('Calibre ebook-convert not found, make sure you specify the path by --calibre /path/to/ebook-convert.')
    return false
}
return true
}

/**
 * @typedef Options
 * @type {object}
 * @property {string} renderer - [node|calibre|wkhtmltopdf] can be either node, calibre and wkhtmltopdf
 * @property {string} calibrePath - path of calibre's ebook-convert
 * @property {string} pdf_size - pdf size limit, in bytes
 * @property {string} white_list - list of file extensions to be included, separate by ','
 * @property {string} format - [mobi|epub|pdf] can be either mobi, epub, pdf
 * @property {string} device - [desktop|tablet|mobile] style can be opt for desktop, tablet and mobile

```

```

* @property {string} baseUrl - base url of CSS style files
* @property {boolean} outline - include PDF outline/bookmarks (default true)
* @property {number} concurrency - max number of parallel Puppeteer render jobs
(default auto)
*/

/**
* Generate ebook from content folder with the given file format
* @param {string} inputFolder - content folder
* @param {string} outputFile - output file name
* @param {string} title - title in ebook file
* @param {Options} options
*/
async function generateEbook(inputFolder, outputFile, title, options = {
  renderer: 'node' }) {
  if (!checkOptions(options)) {
    return
  }

  const { pdf_size, white_list, renderer } = options
  const repoBook = new RepoBook(inputFolder, title, pdf_size, white_list)

  const defaultOutputFileName = getFileName(inputFolder) + '.pdf'
  const outputFileName = outputFile || defaultOutputFileName

  options.outputFileName = outputFileName
  options.inputFolder = inputFolder
  options.outputFile = outputFile

  const outputFiles = []
  while (repoBook.hasNextPart()) {
    const mdString = repoBook.render()
    if (!mdString) {
      console.log('Nothing to generate.')
      return
    }
    let outputFile = null
    outputFile = getHTMLFiles(mdString, repoBook, options)

    if (!outputFile) {
      console.log('Fail to generate HTML files that are used to generate PDFs.')
      break
    }
    outputFiles.push(outputFile)
  }
  if (renderer === 'calibre') {
    sequenceRenderEbook(outputFiles, options)
  } else if (renderer === 'node') {
    await sequenceRenderPDF(outputFiles, options)
  } else if (renderer === 'wkhtmltopdf') {
    renderPDF(outputFiles, options)
  }
}

```



```
    }  
  }  
  
  module.exports = { generateEbook }
```

src/index.js

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```
const { generateEbook } = require('./html')  
module.exports = { generateEbook }
```

src/puppeteer.js

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```
const puppeteer = require('puppeteer');
const pageRenderTimeout = 10;

class HTML2PDF {
  constructor() {
    this.puppeteerConfig = {
      headless: 'new',
      args: [
        '--no-sandbox',
        '--disable-dev-shm-usage',
      ],
    };
  }

  this._initialized = false;
  this._initializing = null;
  this.chrome = null;
}

async _initializePlugins() {
  if (this._initialized) {
    return;
  }

  if (this._initializing) {
    await this._initializing;
    return;
  }

  this._initializing = (async () => {
    this.chrome = await puppeteer.launch(this.puppeteerConfig);
    this._initialized = true;
  })();

  try {
    await this._initializing;
  } finally {
    this._initializing = null;
  }
}

async close() {
  if (this._initializing) {
    await this._initializing;
  }

  if (this.chrome) {
    await this.chrome.close();
  }
  this.chrome = null;
  this._initialized = false;
}
```

```

}

async pdf(templatePath, outputPath, options = {}) {
  await this._initializePlugins();
  const { outline = true } = options;
  const puppeteerPage = await this.chrome.newPage();
  await puppeteerPage.setJavaScriptEnabled(false);

  try {
    await puppeteerPage.goto('file:' + templatePath, {
      waitUntil: 'load',
      timeout: 1000 * (pageRenderTimeout || 30)
    })
    const pdfOptions = {
      path: outputPath,
      displayHeaderFooter: false,
      printBackground: true,
      timeout: 3 * 60 * 1000,
      outline,
    };

    await puppeteerPage.pdf(pdfOptions);
  } finally {
    await puppeteerPage.close();
  }
}

module.exports = new HTML2PDF();

```

src/render.js

[to top](#)

```

const path = require('path')
const fs = require('fs')
const os = require('os')
const { spawnSync } = require('child_process')

const { getFileNameExt } = require('./utils')
const html2PDF = require('./puppeteer');
const wkhtml2PDF = require('./wkhtmltopdf');

let startTs

function removeRelaxedjsTempFiles(outputFileName) {
  const prefix =
outputFileName.split('.').reverse().slice(1).reverse().join('.')
  const html = spawnSync('rm', [`${prefix}.html`])
  const htm = spawnSync('rm', [`${prefix}_temp.htm`])
}

function reportPerformance(outputFileName, startTs) {
  const ts = (Date.now() - startTs) / 1000
  console.log(`${outputFileName} created in ${ts} seconds.`)
}

function getPDFConcurrency(totalFiles, options = {}) {
  if (totalFiles <= 1) {
    return 1
  }

  const requested = Number(options.concurrency)
  if (Number.isInteger(requested) && requested > 0) {
    return Math.min(totalFiles, requested)
  }

  const cpuCount = (os.cpus() || []).length || 1
  // Keep memory usage predictable while still using multiple cores.
  return Math.min(totalFiles, Math.max(1, Math.min(4, cpuCount)))
}

async function sequenceRenderPDF(docFiles, options = {}) {
  const concurrency = getPDFConcurrency(docFiles.length, options)

  try {
    if (concurrency === 1) {
      await docFiles.reduce(
        (promise, file) => promise.then(() => html2PDF.pdf(file,
file.replace('.html', '.pdf'), options)),
        Promise.resolve()
      )
    }
    return
  }
}

```

```

let nextFileIndex = 0
const renderNext = async () => {
  const i = nextFileIndex
  nextFileIndex += 1
  if (i >= docFiles.length) {
    return
  }

  const file = docFiles[i]
  await html2PDF.pdf(file, file.replace('.html', '.pdf'), options)
  await renderNext()
}

const workers = Array.from({ length: concurrency }, () => renderNext())
await Promise.all(workers)
} finally {
  // close chrome so that cli can terminate
  await html2PDF.close()
}
}

```

```

function renderPDF(docFiles, options = {}) {
  const parallel = 1;
  const total = docFiles.length;
  for (let i = 0; i < total; i++) {
    let j = 0;
    for (; j < parallel && (i + j) < total; j++) {
      const file = docFiles[i + j];
      wkhtml2PDF(file, file.replace('.html', '.pdf'), options);
    }
    i += j;
  }
}

```

```

function sequenceRenderEbook(docFiles, options, i = 0) {
  const { outputFileName, renderer, calibrePath, format } = options

  if (i === 0) {
    startTs = Date.now()
  }
  if (i >= docFiles.length) {
    if (renderer === 'node') {
      removeRelaxedjsTempFiles(outputFileName)
    }
    reportPerformance(outputFileName, startTs)
    return
  }
  const docFile = path.resolve(process.cwd(), docFiles[i])
  const formatFile = getFileNameExt(docFile, format)

```

```

const formatArgs = {
  pdf: [
    '--pdf-add-toc',
    '--paper-size',
    'a4',
    '--pdf-default-font-size',
    '12',
    '--pdf-mono-font-size',
    '12',
    '--pdf-page-margin-left',
    '2',
    '--pdf-page-margin-right',
    '2',
    '--pdf-page-margin-top',
    '2',
    '--pdf-page-margin-bottom',
    '2',
    '--page-breaks-before',
    '/',
  ],
  mobi: ['--mobi-toc-at-start', '--output-profile', 'kindle_dx'],
  epub: ['--epub-inline-toc', '--output-profile', 'ipad3', '--flow-size',
'1000'],
}

const args = {
  calibre: [calibrePath, [docFile, formatFile].concat(formatArgs[format])]
}

if (renderer === 'calibre') {
  if (!fs.existsSync(calibrePath)) {
    console.log(`Calibre is not available.`)
  }
}

const cmd = args[renderer]
const res = spawnSync(cmd[0], cmd[1])
if (res.error) {
  console.log(`Some error happened when creating ${formatFile}.`)
  console.error(res.error);
  return
}

if (!fs.existsSync(formatFile)) {
  console.log(`${docFiles[i]} was not rendered. ${formatFile}`)
}

sequenceRenderEbook(docFiles, options, i + 1)
}

```

```
module.exports = { sequenceRenderEbook, sequenceRenderPDF, renderPDF }
```

src/repo.js

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```

const path = require('path')
const fs = require('fs')
const hljs = require('highlight.js')
const { getFileName, getCleanFilename } = require('./utils')

class RepoBook {
  constructor(dir, title, pdf_size, white_list) {
    this.title = title
    this.pdf_size = pdf_size

    this.blackList = ['node_modules', 'vendor']
    this.whiteList = white_list ? white_list.split(',') : null

    this.aliases = {}
    this.byteOffset = 0
    this.fileOffset = 0
    this.partOffset = 0
    this.done = false
    this.dir = dir

    this.files = this.readDir(dir)
    this.registerLanguages()
  }

  hasNextPart() {
    return this.done === false
  }

  hasSingleFile() {
    return this.partOffset === 0 // effective after at least calling render()
    once
  }

  currentPart() {
    return this.partOffset
  }

  registerLanguage(name, language) {
    const lang = language(hljs)
    if (lang && lang.aliases) {
      name = name.split('.')[0]
      if (this.whiteList) {
        if (this.whiteList.indexOf(name) > -1) {
          this.aliases[name] = name
        }
      } else {
        this.aliases[name] = name
      }
    }

    lang.aliases.map(alias => {

```



```

    if (this.whiteList) {
      if (this.whiteList.indexOf(alias) > -1) {
        this.aliases[alias] = name.split('.')[0]
      }
    } else {
      this.aliases[alias] = name.split('.')[0]
    }
    return null
  })
}
}

```

```

registerLanguages() {
  const listPath = path.join(path.dirname(require.resolve('highlight.js')),
    'languages')
  fs.readdirSync(listPath).map(f => {
    this.registerLanguage(f, require(path.join(listPath, f)))
    return null
  })
}

```

```

readDir(dir, allFiles = [], level = 0) {
  const files = fs
    .readdirSync(dir)
    .map(f => path.join(dir, f))
    .filter(f => fs.lstatSync(f).size / 1000 < 2000) // smaller than 2m
    .map(f => [f, level])

```

```

  level > 0 ? allFiles.push([dir, level, true]) : null // push folder name

```

```

  files.map(pair => {
    const f = pair[0]
    const blacklistHit = this.blackList.filter(e => !!f.match(e)).length > 0
    if (!fs.lstatSync(f).isDirectory()) {
      allFiles.push([f, level + 1, false])
    } else {
      path.basename(f)[0] !== '.' && // ignore hidden folders
      blacklistHit === false &&
      this.readDir(f, allFiles, level + 1)
    }
    return null
  })
  return allFiles
}

```

```

renderIndex(files) {
  return files
    .filter(f => {
      const fileName = getFileName(f[0])
      const ext = path.extname(fileName).slice(1)
      const isFolder = fs.lstatSync(f[0]).isDirectory()

```

```

    return fileName[0] !== '.' && (ext in this.aliases || isFolder)
  })
  .map(f => {
    const indexName = getCleanFilename(f[0], this.dir, f[1]),
      anchorName = getCleanFilename(f[0], this.dir),
      left_pad = f[2] ? '&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;' .repeat(f[1]) : '',
      h_level = '####',
      list_style = f[2] ? '' : '&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;&nbsp;' .repeat(f[1]) + '|-'
    ,
      return f[2] ? `${h_level} ${left_pad} ${list_style} /${indexName}` :
        `${h_level} ${left_pad}[${list_style} ${indexName}](#${anchorName})`
    })
    .join('\n')
  }

render() {
  const files = this.files
  const contents = []
  let i = this.fileOffset

  for (; i < files.length; i++) {
    const file = files[i][0]

    const fileName = getFileName(file)
    if (fs.statSync(file).isDirectory()) {
      continue
    }

    const ext = path.extname(fileName).slice(1)

    if (ext.length === 0) {
      continue
    }

    if (fileName[0] === '.') {
      continue
    }

    const lang = this.aliases[ext]
    if (lang) {
      let data = fs.readFileSync(file)
      if (ext === 'md') {
        data = `#### ${getCleanFilename(file, this.dir)} \n[to top]
(#Contents)` + '\n' + data + '\n'
      } else {
        data = `#### ${getCleanFilename(file, this.dir)} \n[to top]
(#Contents)` + '\n` ` ` ` ' + lang + '\n' + data + '\n` ` ` ` \n'
      }
      contents.push(data)

      this.byteOffset += data.length * 2
    }
  }
}

```

```

    if (this.byteOffset > this.pdf_size) {
      // if more than one part
      const title = `# ${this.title} (${++this.partOffset})\n\n\n\n`

      let toc = '## Contents\n'
      const index = this.renderIndex(files.slice(this.fileOffset, i + 1))
      toc += index
      contents.unshift(title, toc)

      if (i === this.fileOffset) {
        // when single file exceeds size limit
        this.fileOffset++
      } else {
        this.fileOffset = i
      }
      this.byteOffset = 0

      return contents.join('\n')
    }
  }
}

if (contents.length === 0) {
  this.done = true
  return null
}

// if one part
const title = this.partOffset
  ? `# ${this.title} (${++this.partOffset})\n\n\n\n` // the last part
  : `# ${this.title} \n\n\n\n` // single pdf

let toc = '## Contents\n'
const index = this.renderIndex(files.slice(this.fileOffset, i + 1))
toc += index
contents.unshift(title, toc)

this.fileOffset = files.length // should return null next round
this.done = true

return contents.join('\n')
}
}

module.exports = RepoBook

```

src/utils.js

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```

const path = require('path')

function getSizeInByte(mb) {
  return mb * 0.8 * 1000 * 1000
}

// '../' => 'untitled'
// './' => 'untitled'
// './path' => 'path'
function getFileName(fpath) {
  const base = path.basename(fpath)
  return base[0] === '.' ? 'untitled' : base
}

function getCleanFilename(filename, folder, depth = 0) {
  filename = filename.replace(folder, '')
  if (folder === '../') {
    depth -= 1
  }
  if (depth > 0) {
    return filename
      .split('/')
      .slice(depth)
      .join('/')
  } else {
    return filename
  }
}

// 'file.random_extension' => 'file.ext'
function getFileNameExt(fileName, ext = 'pdf') {
  return fileName.replace(/\.[0-9a-zA-Z]+$/, `.${ext}`)
}

module.exports = { getSizeInByte, getFileName, getCleanFilename, getFileNameExt }

```

src/wkhtmltopdf.js

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```
const { spawnSync, spawn } = require('child_process');

module.exports = function pdf(templatePath, outputPath, options = {}) {
  const { outline = true } = options;
  const args = [
    '--disable-javascript',
    '--enable-local-file-access',
    '--default-header',
    '--margin-top', '0',
    '--margin-bottom', '0',
    '--allow', './html5bp',
  ];

  if (outline) {
    args.push('--outline');
  }

  args.push(templatePath, outputPath);

  const res = spawnSync('wkhtmltopdf', args);
  if (res.error) {
    console.log('fail to generate pdf using wkhtmltopdf', res.error);
    throw new Error(res.error);
  }
};
```

test/outline.test.js

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```

const fs = require('fs')
const path = require('path')
const { spawnSync } = require('child_process')

const { getPdfOutlineCount } = require('./utils/pdf-outline')

jest.setTimeout(3 * 60 * 1000)

describe('PDF outline/bookmarks', () => {
  const fixtureWithOutline = path.resolve(__dirname, 'fixtures/with-outline.pdf')
  const fixtureWithoutOutline = path.resolve(__dirname, 'fixtures/without-outline.pdf')

  it('detects outline count from PDF bytes', () => {
    expect(getPdfOutlineCount(fixtureWithOutline)).toBeGreaterThan(0)
    expect(getPdfOutlineCount(fixtureWithoutOutline)).toBe(0)
  })

  it('generates PDF bookmarks end-to-end', () => {
    const result = spawnSync(process.execPath, [path.resolve(__dirname, 'scripts/check-outline.js')], {
      encoding: 'utf8',
    })

    if (result.status !== 0) {
      throw new Error(result.stderr || result.stdout || 'outline check failed')
    }

    expect(result.stdout).toMatch(/outline enabled:/)
    expect(result.stdout).toMatch(/outline disabled:/)
  })
})

```

test/performance.test.js

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```

const fs = require('fs')
const path = require('path')
const os = require('os')

jest.mock('../src/puppeteer', () => {
  const fs = require('fs')

  return {
    pdf: jest.fn(async (templatePath, outputPath) => {
      fs.writeFileSync(outputPath, `mock-pdf:${templatePath}`)
    }),
    close: jest.fn(),
  }
})

const { generateEbook } = require('../src/html')
const { getSizeInByte } = require('../src/utils')

const PDF_SIZE = getSizeInByte(3)
const protocol = os.name === 'windows' ? 'file:///': 'file://'
const baseUrl = path.resolve(__dirname, '../html5bp')

jest.setTimeout(2 * 60 * 1000)

describe('render', () => {
  it('with default', async () => {
    const srcPath = './test/data/redis-7.0.0/src'
    if (fs.existsSync(srcPath)) {
      const output = 'redis.pdf'
      await generateEbook(srcPath, output, 'Redis', {
        renderer: 'node',
        calibrePath: '/Applications/calibre.app/Contents/MacOS/ebook-convert',
        pdf_size: PDF_SIZE,
        white_list: null,
        format: 'pdf',
        device: 'desktop',
        baseUrl,
        protocol
      })
      // 6 pdfs with a size limit of 3mb, redis-7.0.0
      expect(fs.existsSync('redis-1.pdf')).toBe(true)
      expect(fs.existsSync('redis-6.pdf')).toBe(true)
    } else {
      console.log('skip because source folder is not specified.')
    }
  })
})

```



```

jest.mock('puppeteer', () => ({
  launch: jest.fn(),
}))

function getPage() {
  return {
    setJavaScriptEnabled: jest.fn().mockResolvedValue(undefined),
    goto: jest.fn().mockResolvedValue(undefined),
    pdf: jest.fn().mockResolvedValue(undefined),
    close: jest.fn().mockResolvedValue(undefined),
  }
}

describe('puppeteer renderer init', () => {
  beforeEach(() => {
    jest.resetModules()
    jest.clearAllMocks()
  })

  it('launches browser only once for concurrent PDF jobs', async () => {
    const puppeteer = require('puppeteer')
    const browser = {
      newPage: jest.fn(async () => getPage()),
      close: jest.fn().mockResolvedValue(undefined),
    }

    puppeteer.launch.mockImplementation(
      () => new Promise((resolve) => setTimeout(() => resolve(browser), 20))
    )

    const html2PDF = require('../src/puppeteer')

    await Promise.all([
      html2PDF.pdf('/tmp/a.html', '/tmp/a.pdf'),
      html2PDF.pdf('/tmp/b.html', '/tmp/b.pdf'),
      html2PDF.pdf('/tmp/c.html', '/tmp/c.pdf'),
    ])

    expect(puppeteer.launch).toHaveBeenCalledTimes(1)
    expect(browser.newPage).toHaveBeenCalledTimes(3)

    await html2PDF.close()
    expect(browser.close).toHaveBeenCalledTimes(1)
  })

  it('passes outline option to page.pdf()', async () => {
    const puppeteer = require('puppeteer')
    const page = getPage()
    const browser = {
      newPage: jest.fn().mockResolvedValue(page),
    }
  })

```

```
        close: jest.fn().mockResolvedValue(undefined),
    }

    puppeteer.launch.mockResolvedValue(browser)

    const html2PDF = require('../src/puppeteer')

    await html2PDF.pdf('/tmp/a.html', '/tmp/a.pdf', { outline: true })
    expect(page.pdf).toHaveBeenCalledWith(
        expect.objectContaining({ outline: true, path: '/tmp/a.pdf' })
    )

    await html2PDF.pdf('/tmp/b.html', '/tmp/b.pdf', { outline: false })
    expect(page.pdf).toHaveBeenCalledWith(
        expect.objectContaining({ outline: false, path: '/tmp/b.pdf' })
    )

    await html2PDF.close()
    })
})
```

test/render-options.test.js

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```

jest.mock('../src/puppeteer', () => ({
  pdf: jest.fn().mockResolvedValue(undefined),
  close: jest.fn(),
}))

const html2PDF = require('../src/puppeteer')
const { sequenceRenderPDF } = require('../src/render')

describe('sequenceRenderPDF options', () => {
  beforeEach(() => {
    jest.clearAllMocks()
  })

  it('passes outline option to puppeteer renderer', async () => {
    await sequenceRenderPDF(['test.html'], { outline: false })

    expect(html2PDF.pdf).toHaveBeenCalled('test.html', 'test.pdf', {
      outline: false })
    expect(html2PDF.close).toHaveBeenCalledTimes(1)
  })

  it('always closes browser when rendering fails', async () => {
    html2PDF.pdf.mockRejectedValueOnce(new Error('render failed'))

    await expect(sequenceRenderPDF(['test.html'], { outline: true
    })).rejects.toThrow('render failed')
    expect(html2PDF.close).toHaveBeenCalledTimes(1)
  })

  it('renders files in parallel when concurrency is set', async () => {
    let inFlight = 0
    let maxInFlight = 0

    html2PDF.pdf.mockImplementation(async () => {
      inFlight += 1
      maxInFlight = Math.max(maxInFlight, inFlight)
      await new Promise((resolve) => setTimeout(resolve, 20))
      inFlight -= 1
    })

    await sequenceRenderPDF(['1.html', '2.html', '3.html'], { concurrency: 2 })

    expect(maxInFlight).toBeGreaterThan(1)
    expect(html2PDF.pdf).toHaveBeenCalledTimes(3)
    expect(html2PDF.close).toHaveBeenCalledTimes(1)
  })
})

```



```

const fs = require('fs')
const path = require('path')
const { spawnSync } = require('child_process')

jest.setTimeout(2 * 60 * 1000)

jest.mock('../src/puppeteer', () => {
  const fs = require('fs')

  return {
    pdf: jest.fn(async (templatePath, outputPath) => {
      fs.writeFileSync(outputPath, `mock-pdf:${templatePath}`)
    }),
    close: jest.fn(),
  }
})

const { generateEbook } = require('../src/html')
const { getSizeInByte } = require('../src/utils')

const PDF_SIZE = getSizeInByte(5) // 10 Mb
const baseUrl = path.resolve(__dirname, '../html5bp')

describe('render', () => {
  it('with default', async () => {
    const output = 'self.pdf'
    await generateEbook('../src', output, 'repo-to-pdf', {
      renderer: 'node',
      calibrePath: '/Applications/calibre.app/Contents/MacOS/ebook-convert',
      pdf_size: PDF_SIZE,
      white_list: null,
      format: 'pdf',
      device: 'desktop',
      baseUrl,
    })
    expect(fs.existsSync(output)).toBe(true)
  })

  it('with mobi', async () => {
    const output = 'self.mobi'
    const calibrePath = '/Applications/calibre.app/Contents/MacOS/ebook-convert'
    if (fs.existsSync(calibrePath)) {
      const check = spawnSync(calibrePath, ['--version'])
      if (check.status !== 0) {
        console.log('skip mobi because calibre is not executable.')
        return
      }
    }

    await generateEbook('../src', output, 'repo-to-pdf', {
      renderer: 'calibre',
    })
  })
})

```

```

        calibrePath,
        pdf_size: PDF_SIZE,
        white_list: null,
        format: 'mobi',
        device: 'desktop',
        baseUrl,
    })

    if (!fs.existsSync(output)) {
        console.log('skip mobi because calibre did not generate output in this
environment.')
        return
    }

    expect(fs.existsSync(output)).toBe(true)
} else {
    console.log('skip mobi because no calibre is found.')
}
})

it('with mobile pdf', async () => {
    const output = 'self-mobile.pdf'
    await generateEbook('./src', output, 'repo-to-pdf', {
        renderer: 'node',
        calibrePath: '/Applications/calibre.app/Contents/MacOS/ebook-convert',
        pdf_size: PDF_SIZE,
        white_list: null,
        format: 'pdf',
        device: 'mobile',
        baseUrl,
    })
    expect(fs.existsSync(output)).toBe(true)
})

it('with tablet pdf', async () => {
    const output = 'self-tablet.pdf'
    await generateEbook('./src', output, 'repo-to-pdf', {
        renderer: 'node',
        calibrePath: '/Applications/calibre.app/Contents/MacOS/ebook-convert',
        pdf_size: PDF_SIZE,
        white_list: null,
        format: 'pdf',
        device: 'tablet',
        baseUrl,
    })
    expect(fs.existsSync(output)).toBe(true)
})

it('with current folder pdf', async () => {
    const output = 'self-current-folder.pdf'
    await generateEbook('./', output, 'repo-to-pdf', {

```

```
    renderer: 'node',
    calibrePath: '/Applications/calibre.app/Contents/MacOS/ebook-convert',
    pdf_size: PDF_SIZE,
    white_list: 'js',
    format: 'pdf',
    device: 'desktop',
    baseUrl,
  })
  expect(fs.existsSync(output)).toBe(true)
})

it('with white list', async () => {
  const output = 'self-whitelist.pdf'
  await generateEbook('./src', output, 'repo-to-pdf', {
    renderer: 'node',
    calibrePath: '/Applications/calibre.app/Contents/MacOS/ebook-convert',
    pdf_size: PDF_SIZE,
    white_list: 'js',
    format: 'pdf',
    device: 'desktop',
    baseUrl,
  })
  expect(fs.existsSync(output)).toBe(true)
})
})
```

test/scripts/check-outline.js

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```

const fs = require('fs')
const path = require('path')
const { spawnSync } = require('child_process')

const { getPdfOutlineCount } = require('../utils/pdf-outline')

const outputDir = path.resolve(__dirname, '../tmp')
const withOutline = path.join(outputDir, 'outline-enabled.pdf')
const withoutOutline = path.join(outputDir, 'outline-disabled.pdf')

function generatePdf(outputFile, outline) {
  const script = `
    const path = require('path')
    const { generateEbook } = require(${JSON.stringify(path.resolve(__dirname,
'../../src/html'))})

    generateEbook('./src', ${JSON.stringify(outputFile)}, 'repo-to-pdf', {
      renderer: 'node',
      pdf_size: 5 * 1000 * 1000,
      white_list: 'js',
      format: 'pdf',
      device: 'desktop',
      baseUrl: ${JSON.stringify(path.resolve(__dirname, '../../html5bp'))},
      outline: ${outline},
    }).then(() => process.exit(0)).catch((error) => {
      console.error(error)
      process.exit(1)
    })
  `

  const result = spawnSync(process.execPath, ['-e', script], {
    cwd: path.resolve(__dirname, '../..'),
    encoding: 'utf8',
  })

  if (result.status !== 0) {
    throw new Error(result.stderr || result.stdout || 'PDF generation failed')
  }
}

function main() {
  fs.mkdirSync(outputDir, { recursive: true })

  generatePdf(withOutline, true)
  generatePdf(withoutOutline, false)

  const enabledCount = getPdfOutlineCount(withOutline)
  const disabledCount = getPdfOutlineCount(withoutOutline)

  if (enabledCount <= 0) {

```



```
const fs = require('fs')

function getPdfOutlineCount(pdfPath) {
  const content = fs.readFileSync(pdfPath).toString('latin1')
  const outlineSection = content.match(/\Type\s*\Outlines[\s\S]*?\Count\s+(-?\d+)/)
  if (!outlineSection) {
    return 0
  }

  return Math.abs(parseInt(outlineSection[1], 10))
}

module.exports = { getPdfOutlineCount }
```